Every isogeny class over $\mathbb{F}_p$, $p = 2^{31} - 1$: an exact census of ECDLP structure

Study `research/fp31_whole_field_census_20261111`, 2026-11-11. Base commit `24690cfeef`. Tool: `tool/` (native Rust, no dependencies), `fp31-class-census census | bruteforce | summary`. Host: 4-vCPU cloud container, no isolation; the run time (211 s) is incidental and not a performance claim. Every number below is an exact computation over the complete field unless it is labelled *model*. No claim here has a pcc claim id: all are **[pcc:unverified]**.

Question and scope

The request was a statistical sweep over a whole field: every curve, every isogeny, every transfer, to see whether ECDLP behaves differently across an isogeny class. Three parts of that question already have answers in this repository or by theorem, so this study measures only the part that has not been done: the structure of **every** isogeny class of one field.

- **Inside one class** (already measured): rho across 38 isogenous curves × 100 seeds, `research/isogeny-class-rho-n45/` (Kruskal–Wallis p = 0.38, 0 of 38 curves flagged); IC relation yield on every curve of full classes at 24, 28 and 32 bits, `research/isogeny_class_ic_yield_20261102/p2_prime_analogues/RESULTS.md` (null supported in 9 of 12 cells, one flag not replicated). By theorem a generic algorithm sees only the group and $\mathrm{Aut}(E)$, and isogenous curves have the same order (`research/isogeny_class_difficulty_20261008/`).
- **Lifting to $\mathbb{F}_{p^2}$** (theorem): $\#E(\mathbb{F}_{p^2}) = (p+1-t)(p+1+t) = \#E \cdot \#E^{\mathrm{tw}}$, and the Weil restriction of $E/\mathbb{F}_{p^2}$ to $\mathbb{F}_p$ is isogenous to $E \times E^{\mathrm{tw}}$. The lift contains the same prime subgroup and adds the twist's; it cannot shrink r. The census confirms it: $r_{p^2} = \max(r, r^{\mathrm{tw}}) \geq r$ in all 185,363 classes (0 exceptions).
- **Covers over a prime field** (theorem): GHS/Weil descent needs a subfield to descend to; $\mathbb{F}_p$ has none. A cover $C \to E$ over $\mathbb{F}_p$ moves the DLP into $\mathrm{Jac}(C)$ over the same $\mathbb{F}_p$, where index calculus for genus $g \geq 3$ costs $\tilde{O}\left(p^{2-\frac{2}{g}}\right)$, more than rho's $p^{\frac{1}{2}}$.

Method

By Tate's theorem each trace t with $|t| \leq 2\sqrt{p}$ is one $\mathbb{F}_p$-isogeny class. For every one of the 185,363 traces $t \in [-92681, 92681]$ the tool computes: $N = p + 1 - t$ and its factorisation; r, the largest prime of N; the twist order $p + 1 + t$ and its largest prime; $D = t^2 - 4p = f^2 D_K$; $h(D_K)$, exactly, by counting reduced forms (`classno.rs`, copied unchanged from `research/isogeny_class_trio_20261017/selector/`); the Hurwitz class number $H(4p - t^2) = \sum_{d \mid f} h(d^2 D_K) \cdot 2/w$, which is the number of $\mathbb{F}_p$-isomorphism classes of curves with trace t (weighted $2/\ |\mathrm{Aut}|$); the embedding degree $k = \mathrm{ord}_{r(p)}$; and the volcano depths $v_{\ell(f)}$. A random curve $y^2 = x^3 + ax + b$ lies in class t with probability $H(4p - t^2)/2p$, which is how "share of curves" is weighted below.

Checks, all passed:

1. $\sum_t H(4p - t^2) = 2p$ exactly: $\sum 6H = 25,769,803,764 = 12p$ (`data/census_p2147483647.jsonl.check.json`).
2. Brute force: every (a, b) over $\mathbb{F}_p$ point-counted naively and binned by trace equals $\frac{p-1}{2} \cdot H$ for every trace at $p \in \{101, 103, 107, 109, 113, 1009, 1013\}$, which covers $p \equiv 1, 5, 7, 11 \pmod{12}$ (and so the $j = 0, 1728$ weights); 0 mismatches, up to 1,025,156 curves per prime.
3. Independent implementation: a PARI prototype (`qfbclassno`, `factor`, `znorder`, via cypari2; not part of the result path, AGENTS.md "no Python") gave the same values in all 185,363 rows on all 17 integer fields.

Results

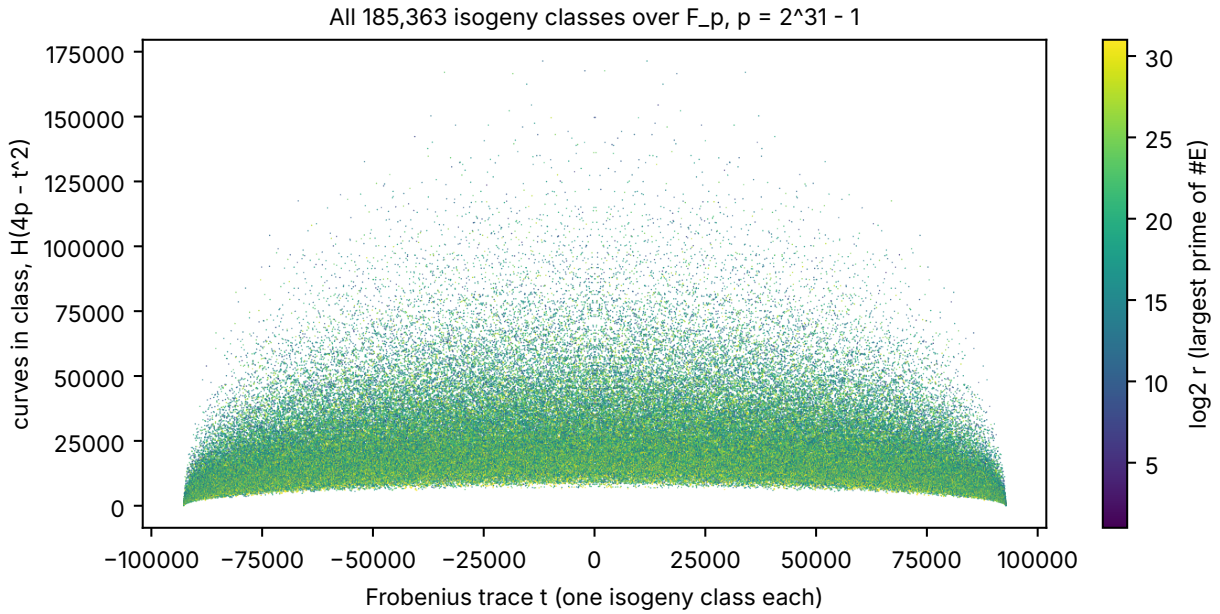


Figure 1: Number of curves $H(4p - t^2)$ in each of the 185,363 isogeny classes against the trace, coloured by $\log_2 r$. The envelope is the Sato–Tate semicircle; the scatter under it is the class number of $t^2 - 4p$. Data: `data/census_p2147483647.jsonl.gz`.

quantity (exact)	**value**
isogeny classes / curves (weighted)	185,363 / 4,294,967,294 = 2p
classes of prime order	8,646 (4.66 % of classes, 2.64 % of curves)
median $\log_2 r$: per curve / per class	18.65 / 19.31
$\log_2 r$ at the 1st percentile of curves	8.07
smallest r	11 ($t = -47584$, $N = 2^5 \cdot 3 \cdot 7^5 \cdot 11^3$, 39,424 curves)
anomalous class ($t = 1$, $N = p$; Smart's attack is polynomial)	17,184 curves
supersingular class ($t = 0$, $N = 2^{31}$)	39,730 curves
smallest embedding degree with $r \geq 2^{20}$	$k = 31$ ($t = -1$, $r = (2^{31} + 1)/3$)
classes where the $\mathbb{F}_{p^2}$ lift has smaller r	0 of 185,363
largest class	171,048 curves ($t = 11752$, $f = 6$)
most endomorphism-ring strata	24 orders ($t = 80012$, $f = 1386$)

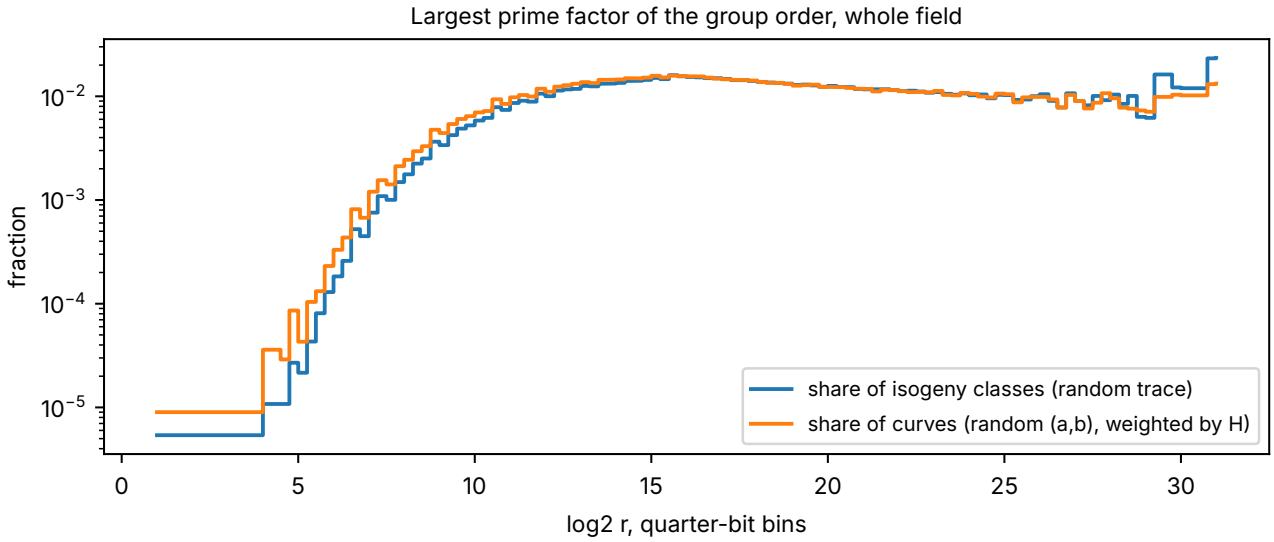


Figure 2: Distribution of $\log_2 r$ over classes (each trace once) and over curves (weighted by H). Recorded bins: `data/summary_p2147483647.json`, `log2_r_hist_quarter_bits`.

The one cross-class difference that exists, and its cause

Class size and smoothness of N are correlated across classes. Spearman correlation over the 185,362 ordinary classes of $\log_2 r$ against: $\log_2 H$, -0.141; number of endomorphism rings, -0.134; $\log_2 f$, -0.122; $\log_2|D_K|$, $+0.104$; $|t|$, -0.0002. Curve-weighted mean $\log_2 r$ falls from 20.34 in classes with one stratum (crater only) to 18.96 with two and 16.35 with eight or more.

This is a known theorem, not a new weakness. If a prime ℓ divides N then $t \equiv p+1 \pmod{\ell}$, so $t^2 - 4p \equiv (p-1)^2$ is a square mod ℓ; small split primes raise $L(1,\chi_D)$ and with it the class number. Equivalently, Frobenius mod ℓ is equidistributed among matrices of determinant p in $\mathrm{GL}_2(\mathbb{F}_\ell)$, so a random curve has $\ell \mid N$ with probability $\ell/(\ell^2-1)$ when $p \equiv 1 \pmod{\ell}$ and $1/(\ell-1)$ otherwise (Lenstra 1987; Howe 1993), while a random trace has $\ell \mid N$ with probability $1/\ell$. The census reproduces both for every prime $\ell \le 47$ with relative error at most 1.3×10^{-4} (figure below, `lenstra_divisibility`).

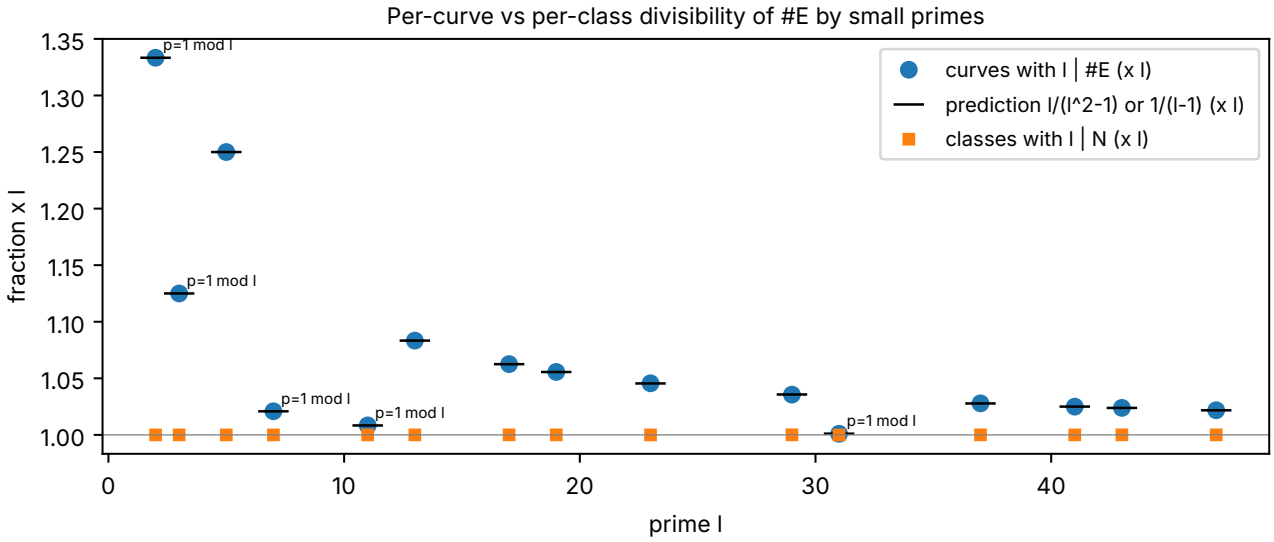


Figure 3: Fraction of curves (circles) and of classes (squares) with $\ell \mid \#E$, times ℓ, against the $\mathrm{GL}_2(\mathbb{F}_\ell)$ prediction (bars). $p = 2^{31} - 1 \equiv 1 \bmod 2, 3, 7, 11, 31$.

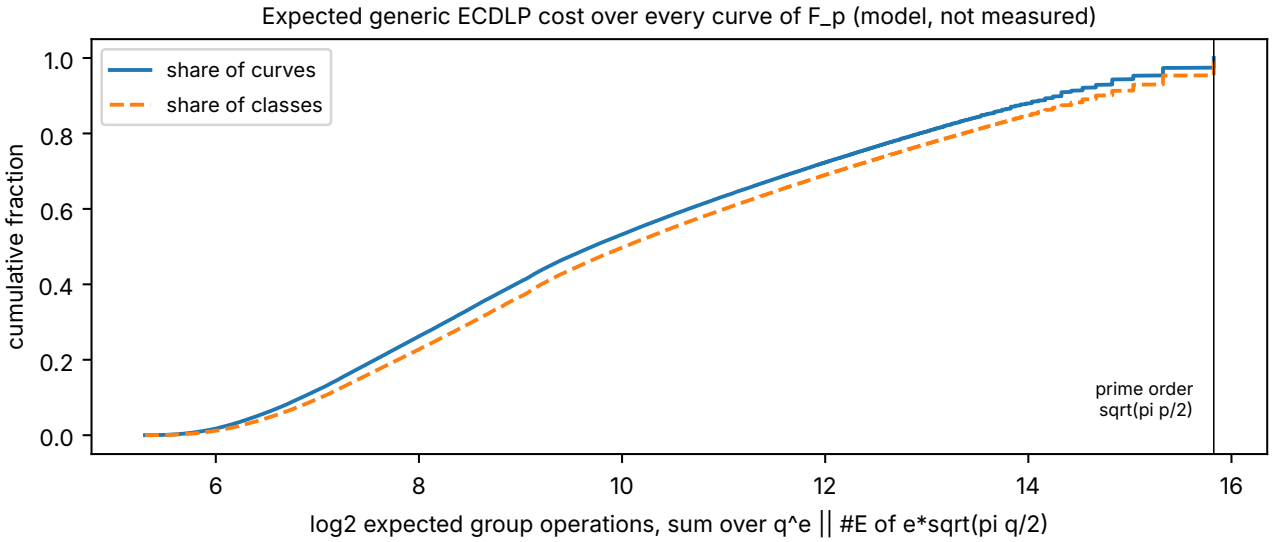


Figure 4: *Model, not measured:* expected generic cost $\sum_{q^e \parallel N} e\sqrt{\pi q/2}$ (Pohlig–Hellman with rho per prime), cumulative over curves and over classes. The median curve needs about $2^{9.7}$ group operations, against $2^{15.8}$ for a curve of prime order. The rows: `rho_cost_ph`.

What this does and does not establish

- Established (exact): the full distribution of group structure, class size, conductor, embedding degree and twist over every isogeny class of $\mathbb{F}_{2^{31}-1}$, and that the only systematic cross-class effect in these invariants is the Lenstra divisibility bias. Every weak class found here is weak for a textbook reason: a small largest prime (Pohlig–Hellman), $t = 1$ (anomalous) or $t = 0$ (supersingular). No class with $r \geq 2^{20}$ has embedding degree below 31.
- Not established: anything about running time. The cost column is a model. A measured check that rho cost is $\sqrt{\pi r/2}$ in every class and depends on nothing else (class size, conductor, D_K) needs a preregistered ecbench run with a benchmark record; it is the proposed next step (README). It leaves the index-calculus baseline unchanged and adds no leaderboard row.